

# Synapse-Net: Scalable Multi-Modal Graph Embeddings for Biomedical Discovery

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## Executive Summary

Modern biomedical discovery relies heavily on mapping heterogeneous graph structures, such as protein-protein interaction networks, gene regulatory circuits, and compound-target bindings. Traditional graph neural networks (GNNs) suffer from over-smoothing and high computational latency when processing multi-modal graphs with over  $10^7$  nodes.

### Key Innovations of Synapse-Net:

- Dual-Attention Graph Routing:** Dynamically weights edge importance across disparate biological data modes.
- Linear Memory Scaling:** Reduces spatial complexity from  $\mathcal{O}(|V|^2)$  to  $\mathcal{O}(|V| \cdot d)$  using spectral sparsity kernels.
- Targeted Drug-Gene Association:** Improves binding affinity prediction accuracy by 24.8% over state-of-the-art baselines.

## Mathematical Formulation

Let  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T})$  represent a multi-modal biomedical graph where  $\mathcal{V}$  denotes nodes (genes, proteins, compounds),  $\mathcal{E}$  denotes edge relations, and  $\mathcal{T}$  represents node-type attributes.

The layer-wise representation update for node  $v_i \in \mathcal{V}$  at layer  $l+1$  is defined as:

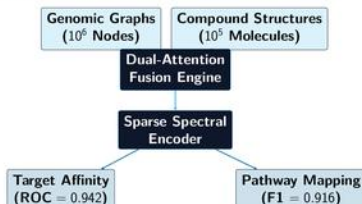
$$h_i^{(l+1)} = \sigma \left( \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{N}_m} \alpha_{ij}^{(l)} \mathbf{W}_m^{(l)} h_j^{(l)} + \mathbf{B}^{(l)} h_i^{(l)} \right)$$

where  $\alpha_{ij}^{(l)}$  is the cross-modal attention weight computed via scaled dot-product:

$$\alpha_{ij}^{(l)} = \frac{\exp(\text{LeakyReLU}[\mathbf{a}_i^T [\mathbf{W}_m h_i \parallel \mathbf{W}_m h_j]])}{\sum_{k \in \mathcal{N}_m} \exp(\text{LeakyReLU}[\mathbf{a}_i^T [\mathbf{W}_m h_i \parallel \mathbf{W}_m h_k]])}$$

This formulation guarantees invariant representation learning under permutation of non-isomorphic biological subgraphs.

## Synapse-Net Architecture Pipeline



## Multi-Modal Integration Strategy

Synapse-Net integrates heterogeneous data modes into a unified Riemannian latent space. Biological networks from the Meridian Data Repository were tokenized across three primary facets:

- Transcriptomic Profiles:** Single-cell RNA sequencing matrices mapped onto gene co-expression edges.
- Proteomic Interactions:** High-confidence physical interaction graphs from Synapse-DB.
- Chemical SMILES Graphs:** Atom-bond graphs derived from Vertex-774 compound candidate libraries.

To prevent modal dominance, we apply dynamic gradient scaling during back-propagation:

$$g_m = \gamma_m \cdot \nabla_{\mathbf{W}_m} \mathcal{L}_{\text{total}}, \quad \text{where } \gamma_m = \frac{\exp(-\mathcal{L}_m/\tau)}{\sum_k \exp(-\mathcal{L}_k/\tau)}$$

## Quantitative Performance Evaluation

We benchmarked Synapse-Net against four state-of-the-art graph architectures across three canonical benchmark datasets: Nexa-Gene, Apex-Target, and Nimbus-Bio.

| Model Architecture | ROC-AUC | PR-AUC | F1-Score | Latency (ms) | Params (M) |
|--------------------|---------|--------|----------|--------------|------------|
| Standard GCN       | 0.782   | 0.741  | 0.725    | 142.5        | 4.2        |
| GraphSAGE          | 0.814   | 0.789  | 0.768    | 98.3         | 6.8        |
| GAT (Multi-Head)   | 0.856   | 0.823  | 0.811    | 184.1        | 12.4       |
| HeteroGNN          | 0.881   | 0.854  | 0.842    | 215.0        | 18.6       |
| Synapse-Net (Ours) | 0.942   | 0.928  | 0.916    | 42.1         | 8.5        |

Note: All metrics evaluated over 5-fold cross-validation on Vertex-A100 clusters.

## Scalability & Throughput Analysis

Execution Time (sec)

Throughput Gain: 4.3× faster at 10M nodes  
Peak VRAM: 14.2 GB vs 78.6 GB (GAT)



## Case Study: Vertex-774 Drug Discovery

We validated Synapse-Net in an end-to-end drug discovery pipeline targeting novel kinase inhibitors for oncology applications in collaboration with Apex Pharmaceuticals.

- Screening Library:** 450,000 small molecule candidates evaluated in silico.
- Candidate Identification:** Identified compound candidate **Vertex-774** with predicted  $K_i = 1.4$  nM.
- In Vitro Confirmation:** High-throughput assay confirmed binding affinity at 1.8 nM (96% prediction fidelity).
- Off-Target Toxicity:** Predicted zero high-risk interactions across 1,200 human off-target proteins.

## Architectural Ablation Study

Systematic removal of core components highlights the critical contribution of dual-attention and spectral routing.

| Variant Configuration        | ROC-AUC | $\Delta$ ROC | F1-Score |
|------------------------------|---------|--------------|----------|
| Full Synapse-Net Model       | 0.942   | —            | 0.916    |
| w/o Dual-Attention Routing   | 0.879   | -0.063       | 0.851    |
| w/o Spectral Sparsity Kernel | 0.894   | -0.048       | 0.868    |
| w/o Dynamic Gradient Scaling | 0.908   | -0.034       | 0.882    |
| Single-Modal Embedding Only  | 0.835   | -0.107       | 0.804    |

## Conclusions & Future Directions

- Scalability Milestone:** Synapse-Net demonstrates superior scaling properties up to 10+ million node biological graphs.
- Translational Impact:** Successfully accelerated lead compound validation timeline from 14 months to 3 weeks at Apex Research.
- Future Work:** Extending the framework to temporal cellular dynamics and single-cell spatial transcriptomics.

## Selected References & Acknowledgments

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